Balanced Three-Phase Circuits

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Structure

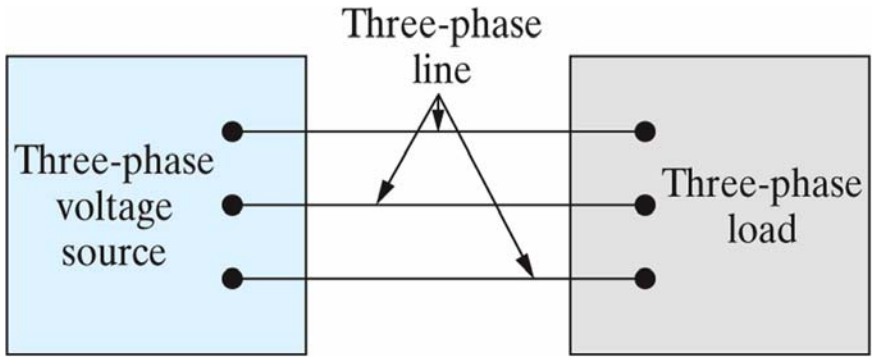
- Balanced Three-Phase Voltages
- Three-Phase Voltage Sources
- Analysis of the Wye-Wye Circuit
- Analysis of the Wye-Delta Circuit
- Power Calculations in Balanced Three-Phase Circuits
- Measuring Average Power in Three-Phase Circuits

Balanced Three-Phase Voltages

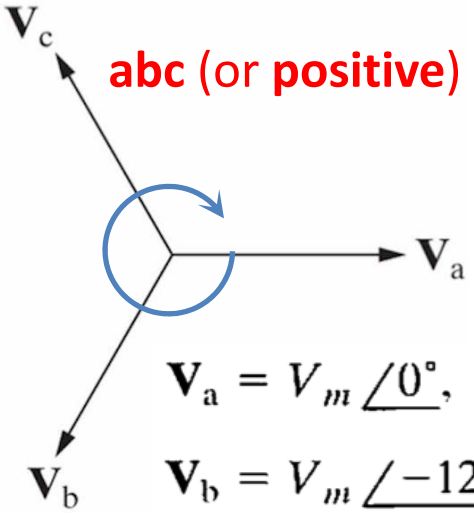
a-phase voltage

b-phase voltage

c-phase voltage



abc (or **positive**) **phase sequence**

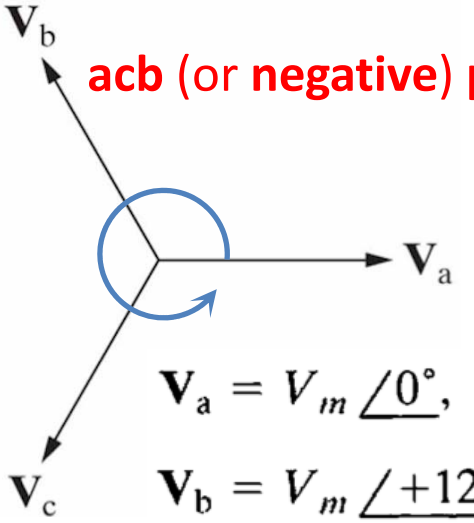


$$\mathbf{V}_a = V_m \angle 0^\circ,$$

$$\mathbf{V}_b = V_m \angle -120^\circ,$$

$$\mathbf{V}_c = V_m \angle +120^\circ,$$

acb (or **negative**) **phase sequence**



$$\mathbf{V}_a = V_m \angle 0^\circ,$$

$$\mathbf{V}_b = V_m \angle +120^\circ,$$

$$\mathbf{V}_c = V_m \angle -120^\circ.$$

The fact that a three-phase circuit can have one of two phase sequences must be taken into account whenever two such circuits operate in parallel. *The circuits can operate in parallel only if they have the same phase sequence*.

If we know the phase sequence and one voltage in the set, we know the entire set.

From Eqs. 11.1 and 11.2, we have

$$\mathbf{V}_a + \mathbf{V}_b + \mathbf{V}_c = 0$$



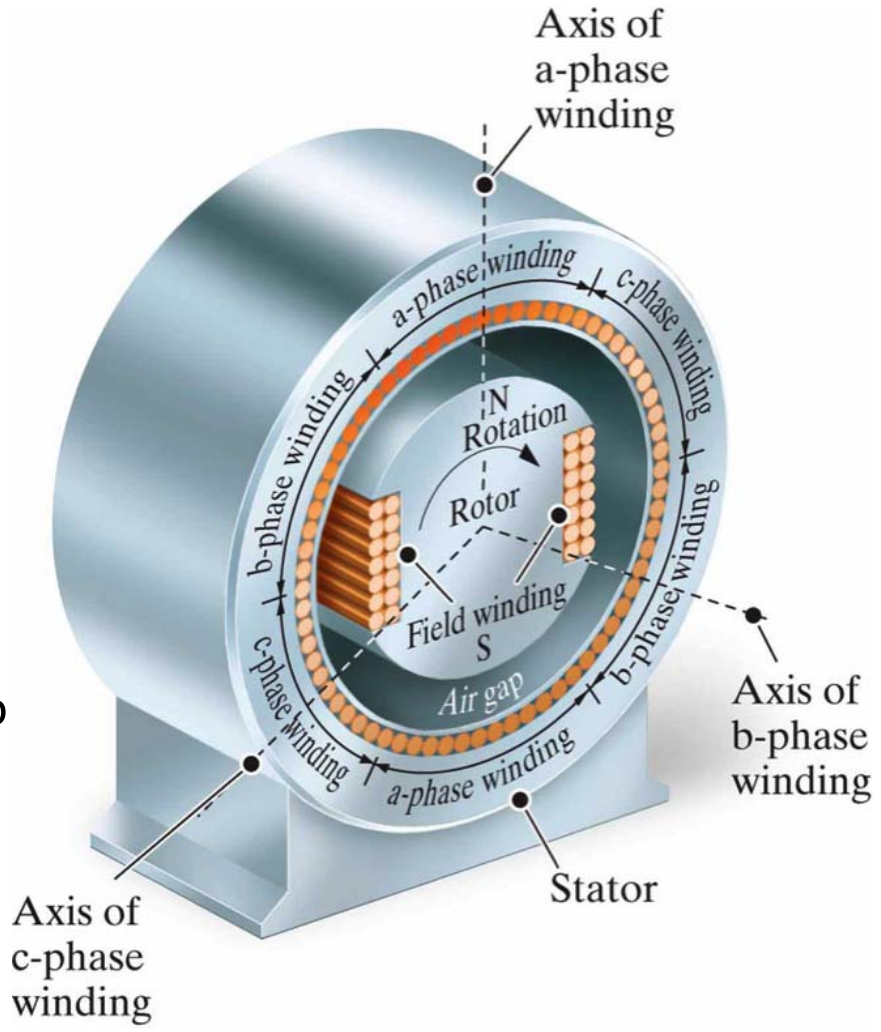
$$v_a + v_b + v_c = 0$$

Three-Phase Voltage Sources

a. *Rotation of the electromagnet induces a sinusoidal voltage in each winding.*

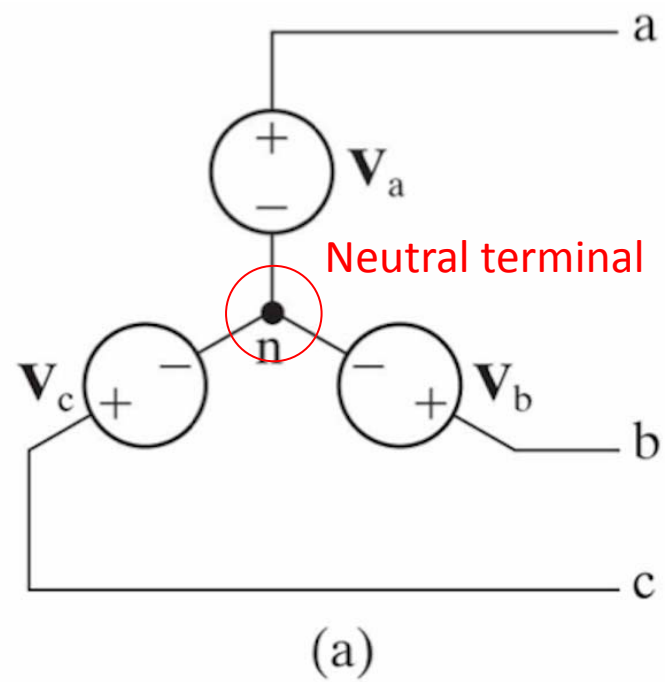
b. The phase windings are designed so that *the sinusoidal voltages induced in them are equal in amplitude and out of phase with each other by 120°*.

c. The phase windings are stationary with respect to the rotating electromagnet, so *the frequency of the voltage induced in each winding is the same*.



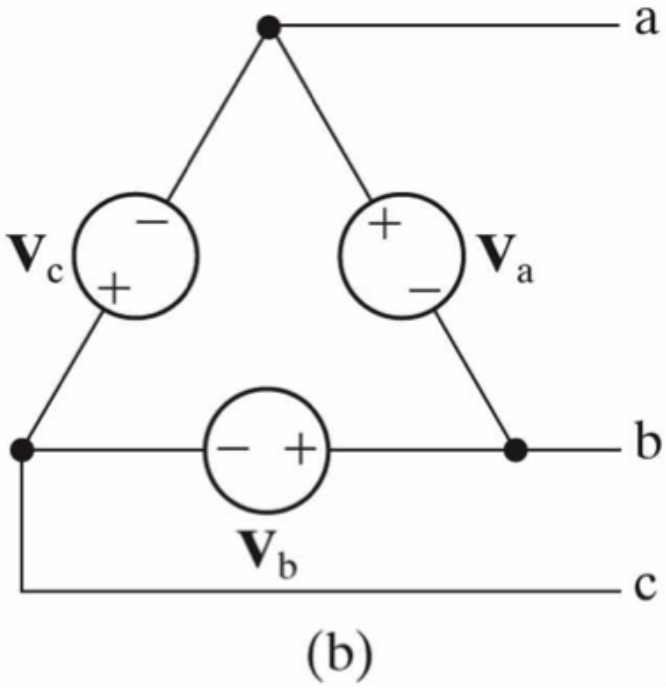
There are two ways of interconnecting the separate phase windings to form a three-phase source:

Y-connected



(a)

Δ -connected



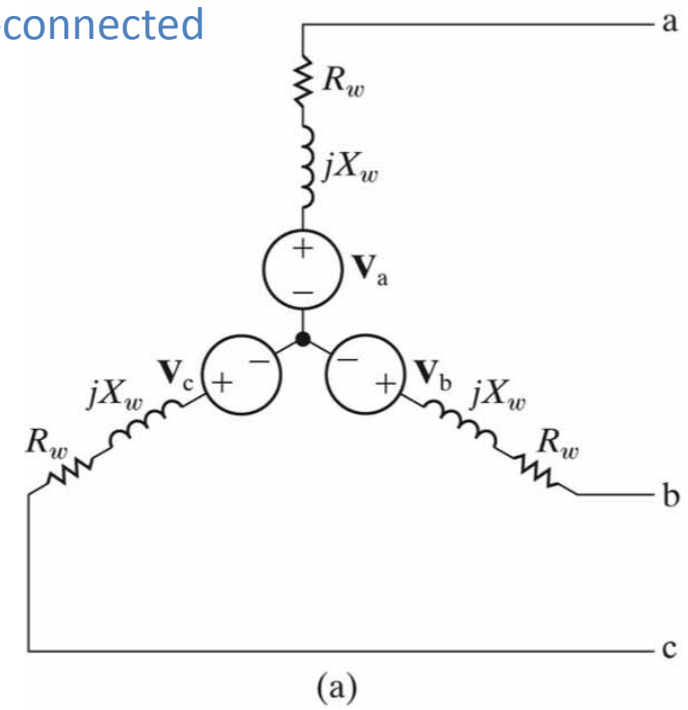
(b)

The neutral terminal *may or may not* be available for external connections.

Here, the model consists solely of *ideal voltage sources*, because the impedance of each phase winding is so small (compared with other impedances in the circuit) that we need not account for it in modeling the generator.

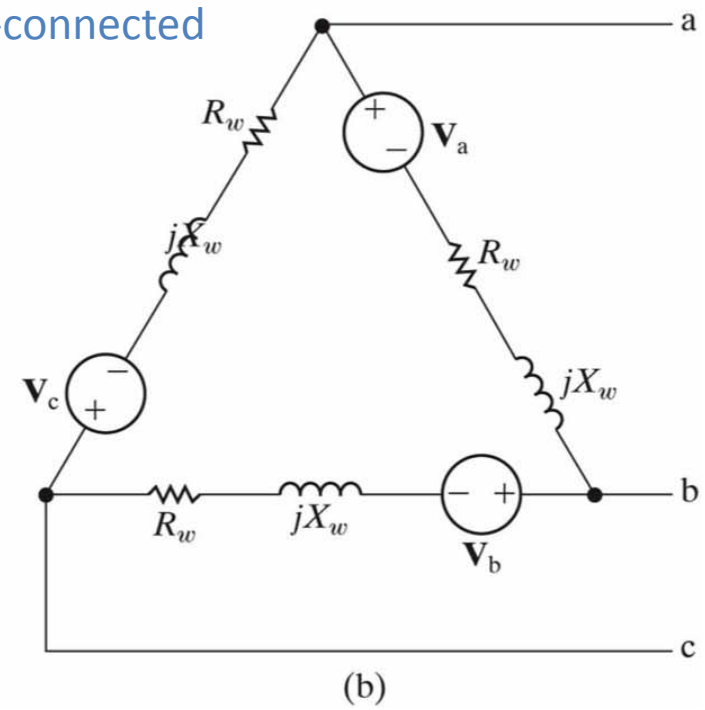
If the impedance of each phase winding is not negligible, we place the *winding impedance in series* with an Ideal sinusoidal voltage source:

Y-connected



(a)

Δ -connected



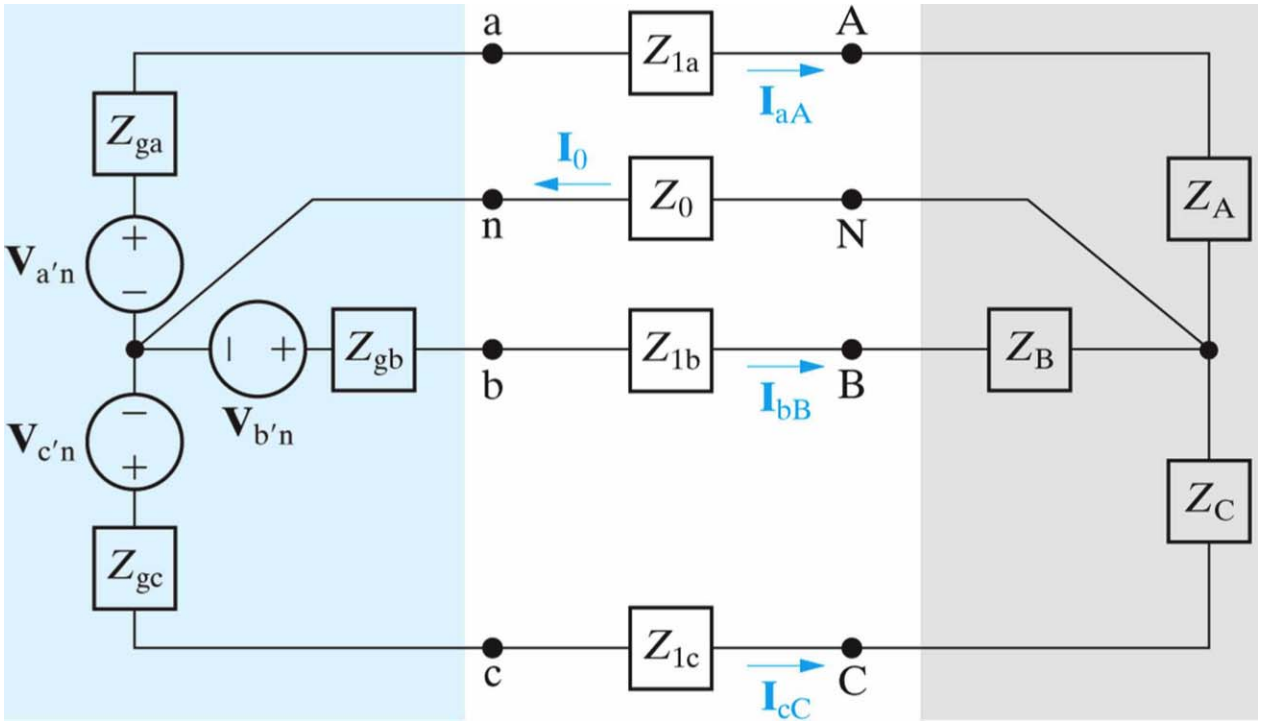
(b)

Four kinds of circuits

Source	Load
Y	Y
Y	Δ
Δ	Y
Δ	Δ

Problems

Analysis of the Wye-Wye Circuit



Internal impedance Line impedance Load impedance

Node-voltage equation:

$$\frac{\mathbf{V}_N}{Z_0} + \frac{\mathbf{V}_N - \mathbf{V}_{a'n}}{Z_A + Z_{1a} + Z_{ga}} + \frac{\mathbf{V}_N - \mathbf{V}_{b'n}}{Z_B + Z_{1b} + Z_{gb}} + \frac{\mathbf{V}_N - \mathbf{V}_{c'n}}{Z_C + Z_{1c} + Z_{gc}} = 0$$

Balanced Three-phase Circuit

a. The voltage sources form a set of balanced three-phase voltages. This means that $\mathbf{V}_{a'n}$, $\mathbf{V}_{b'n}$, and $\mathbf{V}_{c'n}$ are a set of balanced three-phase voltages.
b. The impedance of each phase of the voltage source is the same. This means that $Z_{ga} = Z_{gb} = Z_{gc}$.
c. The impedance of each line (or phase) conductor is the same. This means that $Z_{1a} = Z_{1b} = Z_{1c}$.
d. The impedance of each phase of the load is the same. This means that $Z_A = Z_B = Z_Q$.

There is no restriction on the impedance of a neutral conductor; its value has no effect on whether the system is balanced.

Then, for a balanced three-phase circuit, Eq. 11.5 can be rewritten as:

$$\mathbf{V}_N\left(\frac{1}{Z_0}+\frac{3}{Z_\phi}\right)=\frac{\mathbf{V}_{a'n}+\mathbf{V}_{b'n}+\mathbf{V}_{c'n}}{Z_\phi}$$

where

$$Z_\phi = Z_A + Z_{la} + Z_{ga} = Z_B + Z_{lb} + Z_{gb} = Z_C + Z_{lc} + Z_{gc}$$

$$\mathbf{V}_N = 0$$

There is no difference in potential between the source neutral, n, and the load neutral, N; consequently, the current in the neutral conductor is zero. Hence *we may either remove the neutral conductor from a balanced Y-Y configuration ($\boldsymbol{I}_0 = 0$) or replace it with a perfect short circuit between the nodes n and N ($\boldsymbol{V}_N = 0$)*.

$$\mathbf{I}_{aA}=\frac{\mathbf{V}_{a'n}-\mathbf{V}_N}{Z_A+Z_{la}+Z_{ga}}=\frac{\mathbf{V}_{a'n}}{Z_\phi}$$

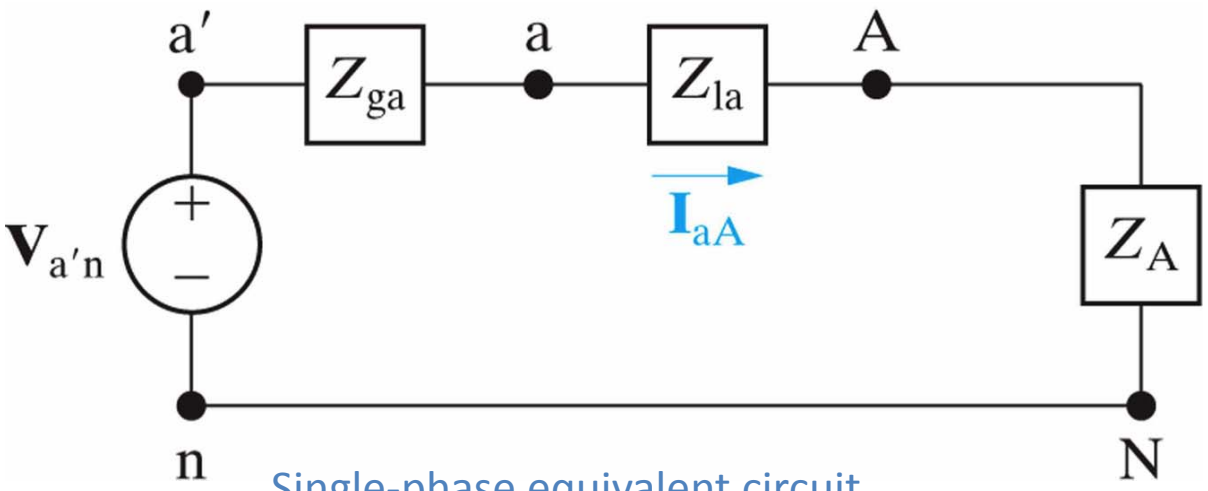
$$\mathbf{I}_{bB}=\frac{\mathbf{V}_{b'n}-\mathbf{V}_N}{Z_B+Z_{lb}+Z_{gb}}=\frac{\mathbf{V}_{b'n}}{Z_\phi}$$

$$\mathbf{I}_{cC}=\frac{\mathbf{V}_{c'n}-\mathbf{V}_N}{Z_C+Z_{lc}+Z_{gc}}=\frac{\mathbf{V}_{c'n}}{Z_\phi}$$

The current in each line is equal in amplitude and frequency and is $120°$ out of phase with the other two line currents. Thus, *if we calculate the current $\boldsymbol{I}_{aA}$ and we know the phase sequence, we have a shortcut for finding $\boldsymbol{I}_{bB}$ and $\boldsymbol{I}_{cC}$.*

Single-phase Equivalent Circuit

Once we know the line current in the figure, calculating any voltages of interest is relatively simple.



Single-phase equivalent circuit

The current in the neutral conductor this circuit is $\mathbf{I}_{aA}$, which is not the same as the current in the neutral conductor of the balanced three-phase circuit, which is

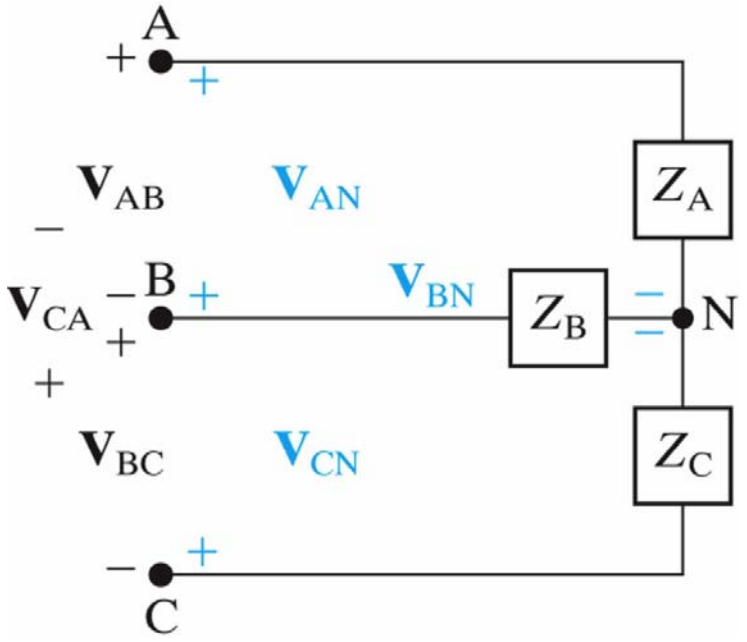
$$\mathbf{I}_o = \mathbf{I}_{aA} + \mathbf{I}_{bB} + \mathbf{I}_{cC} = 0$$

Balanced

Of particular interest is *the relationship between the line-to-line voltages and the line-to-neutral voltages*.

$$\mathbf{V}_{AB} = \mathbf{V}_{AN} - \mathbf{V}_{BN}$$
$$\mathbf{V}_{BC} = \mathbf{V}_{BN} - \mathbf{V}_{CN}$$
$$\mathbf{V}_{CA} = \mathbf{V}_{CN} - \mathbf{V}_{AN}$$

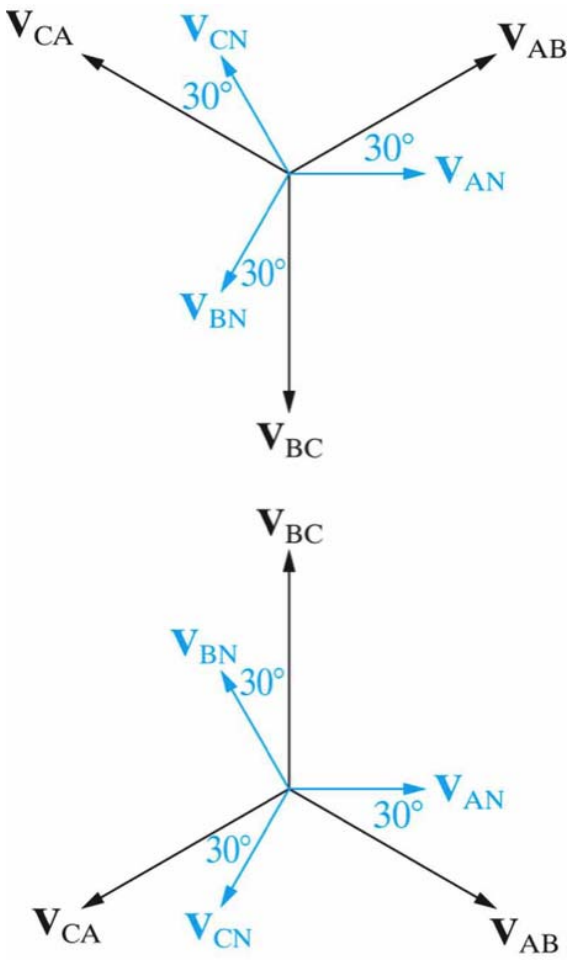
We assume a positive, or abc, sequence and the line-to-neutral voltage of the a-phase as the reference:



$$\mathbf{V}_{AN} = V_\phi \angle 0^\circ$$
$$\mathbf{V}_{CN} = V_\phi \angle +120^\circ$$
$$\mathbf{V}_{BN} = V_\phi \angle -120^\circ$$

$$\mathbf{V}_{AB} = V_\phi \angle 0^\circ - V_\phi \angle -120^\circ = \sqrt{3} V_\phi \angle 30^\circ$$
$$\mathbf{V}_{BC} = V_\phi \angle -120^\circ - V_\phi \angle 120^\circ = \sqrt{3} V_\phi \angle -90^\circ$$
$$\mathbf{V}_{CA} = V_\phi \angle 120^\circ - V_\phi \angle 0^\circ = \sqrt{3} V_\phi \angle 150^\circ$$

1. The magnitude of the line-to-line voltage is $\sqrt{3}$ times the magnitude of the line-to-neutral voltage.
2. The line-to-line voltages form a balanced three-phase set of voltages.
3. The set of line-to-line voltages leads the set of line-to-neutral voltages by $30°$.



Line voltage: the voltage across any pair of lines
Phase voltage: the voltage across a single phase
Line current: the current in a single line
Phase current: current in a single phase

Observe that *in a Δ connection, line voltage and phase voltage are identical*, and *in a **Y** connection, line current and phase current are identical*.

Some Notes

- Because three-phase systems are designed to handle large blocks of electric power, *all voltage and current specifications are given as rms values*.
- *When voltage ratings are given, they refer specifically to the rating of the line voltage*. Thus when a three-phase transmission line is rated at $345\ \text{kV}$, the nominal value of the rms line-to-line voltage is $345{,}000\ \text{V}$.
- *In this chapter we express all voltages and currents as rms values*.
- Finally, the Greek letter phi (ϕ) is widely used in the literature to denote a per-phase quantity. Thus $\mathbf{V}_\phi$, $\mathbf{I}_\phi$, Z_ϕ, P_ϕ and Q_ϕ are interpreted as voltage/phase, current/phase, impedance/phase, power/phase, and reactive power/phase, respectively.

Example #1

A balanced three-phase Y-connected generator with positive sequence has an impedance of $0.2 + j0.5\ \Omega/\phi$ and an internal voltage of 120 V/ϕ. The generator feeds a balanced three-phase Y-connected load having an impedance of $39 + j28\ \Omega/\phi$. The impedance of the line connecting the generator to the load is $0.8 + j1.5\ \Omega/\phi$. The a-phase internal voltage of the generator is specified as the reference phasor.

- Construct the a-phase equivalent circuit of the system.
- Calculate the three line currents $\mathbf{I}_{aA}$, $\mathbf{I}_{bB}$, and $\mathbf{I}_{cC}$.

a) Calculate the three phase voltages at the load. $\mathbf{V}_{AN}$, $\mathbf{V}_{BN}$, and $\mathbf{V}_{CN}$.

b) Calculate the line voltages $\mathbf{V}_{AB}$, $\mathbf{V}_{BC}$, and $\mathbf{V}_{CA}$ at the terminals of the load.

c) Calculate the phase voltages at the terminals of the generator, $\mathbf{V}_{an}$, $\mathbf{V}_{bn}$, and $\mathbf{V}_{cn}$.

d) Calculate the line voltages $\mathbf{V}_{ab}$, $\mathbf{V}_{bc}$, and $\mathbf{V}_{ca}$ at the terminals of the generator.

Solution for Example #1

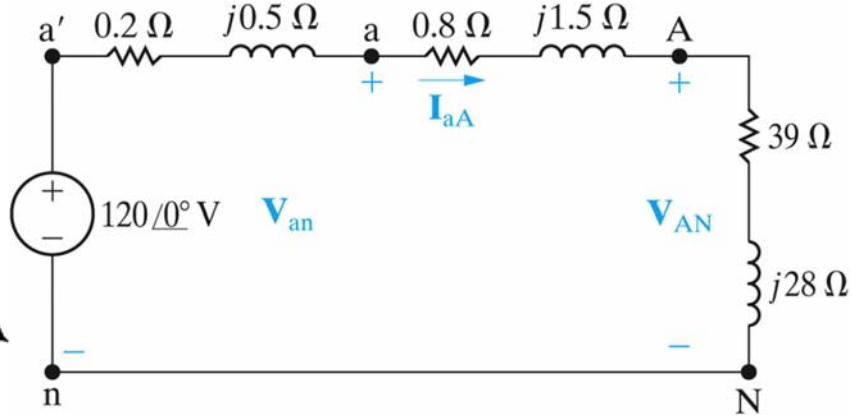
b)

$$\mathbf{I}_{aA} = \frac{120\angle 0^\circ}{(0.2 + 0.8 + 39) + j(0.5 + 1.5 + 28)}$$

$$= \frac{120\angle 0^\circ}{40 + j30} = 2.4\angle -36.87^\circ \text{ A}$$

$$\mathbf{I}_{bB} = 2.4\angle -156.87^\circ \text{ A} \qquad \mathbf{I}_{cC} = 2.4\angle 83.13^\circ \text{ A}$$

a)



c)

$$\mathbf{V}_{AN} = (39 + j28)(2.4\angle -36.87^\circ) = 115.22\angle -1.19^\circ \text{ V}$$

$$\mathbf{V}_{BN} = 115.22\angle -121.19^\circ \text{ V} \qquad \mathbf{V}_{CN} = 115.22\angle 118.81^\circ \text{ V}$$

d)

$$\mathbf{V}_{AB} = (\sqrt{3}\angle 30^\circ)\mathbf{V}_{AN} = 199.58\angle 28.81^\circ \text{ V}$$

$$\mathbf{V}_{BC} = 199.58\angle -91.19^\circ \text{ V} \qquad \mathbf{V}_{CA} = 199.58\angle 148.81^\circ \text{ V}$$

e)

$$\mathbf{V}_{an} = 120 - (0.2 + j0.5)(2.4 \angle -36.87^\circ)$$

$$= 120 - 1.29 \angle 31.33^\circ = 118.90 - j0.67 = 118.90 \angle -0.32^\circ \text{ V}$$

$$\mathbf{V}_{bn} = 118.90 \angle -120.32^\circ \text{ V} \qquad \mathbf{V}_{cn} = 118.90 \angle 119.68^\circ \text{ V}$$

f)

$$\mathbf{V}_{ab} = (\sqrt{3} \angle 30^\circ)\mathbf{V}_{an} = 205.94 \angle 29.68^\circ \text{ V}$$

$$\mathbf{V}_{bc} = 205.94 \angle -90.32^\circ \text{ V} \qquad \mathbf{V}_{ca} = 205.94 \angle 149.68^\circ \text{ V}$$

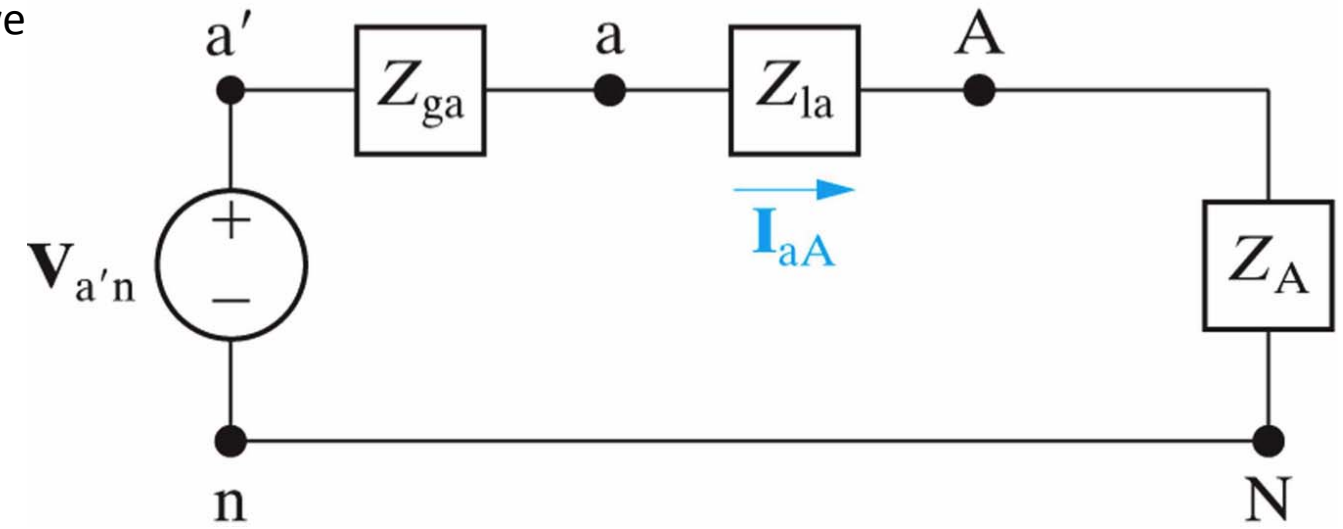
Analysis of the Wye-Delta Circuit

If the load in a three-phase circuit is connected in a delta, it can be transformed into a wye by using the delta-to-wye transformation discussed in Section 9.6. *When the load is balanced, the impedance of each leg of the wye is one third the impedance of each leg of the delta, or*

For *balanced* load, we have

$$Z_Y = \frac{Z_\Delta}{3}$$

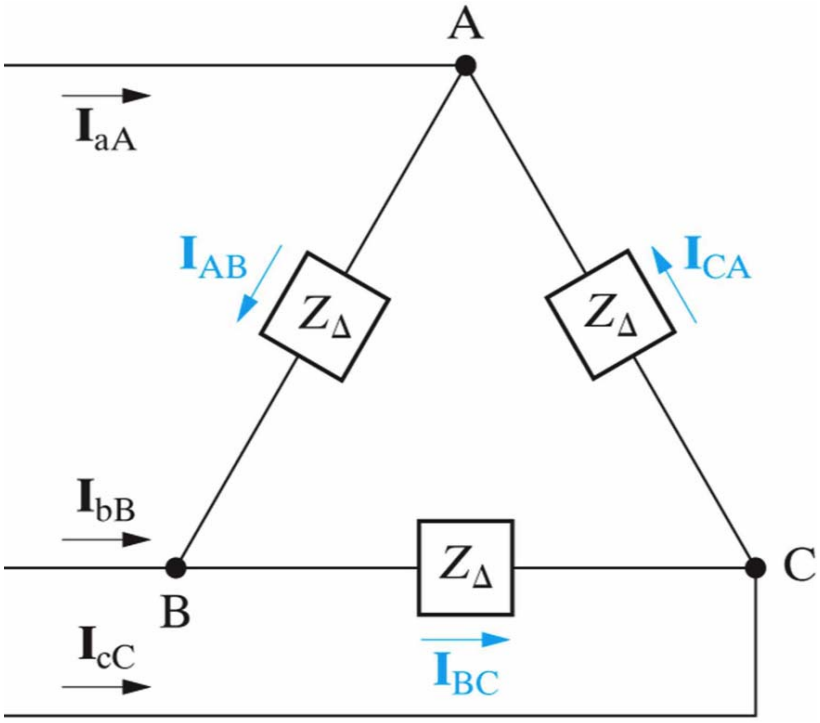
See Eqs. 9.51-9.53



We use this circuit to calculate the line currents, and we then use the line currents to find the currents in each leg of the original Δ load. The relationship between the line currents and the currents in each leg of the delta can be derived using the circuit shown in the Figure.

To demonstrate the relationship between the phase currents and line currents, *we assume a positive phase sequence and let I_ϕ represent the magnitude of the phase current*. Then

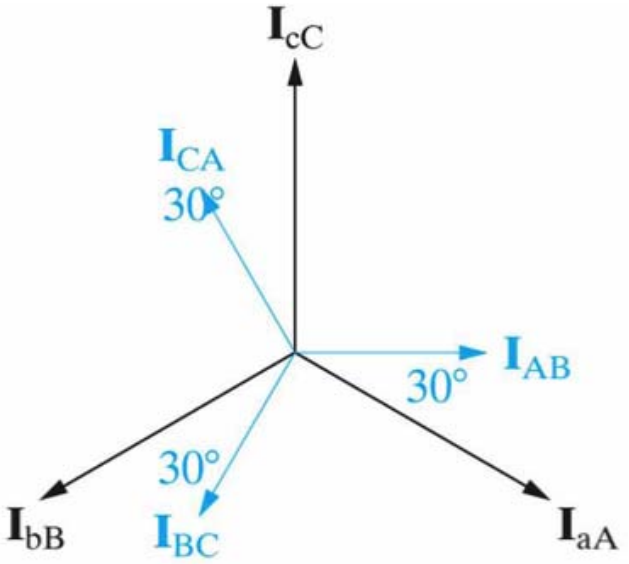
$$\mathbf{I}_{AB} = I_\phi \angle 0^\circ$$
$$\mathbf{I}_{BC} = I_\phi \angle -120^\circ$$
$$\mathbf{I}_{CA} = I_\phi \angle 120^\circ$$



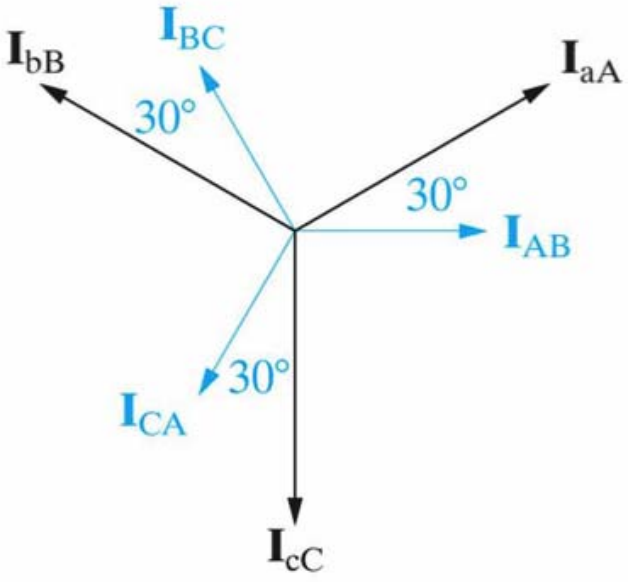
$$\mathbf{I}_{aA} = \mathbf{I}_{AB} - \mathbf{I}_{CA} = I_\phi \angle 0^\circ - I_\phi \angle 120^\circ$$
$$= \sqrt{3} I_\phi \angle -30^\circ$$

$$\mathbf{I}_{bB} = \mathbf{I}_{BC} - \mathbf{I}_{AB} = I_\phi \angle -120^\circ - I_\phi \angle 0^\circ$$
$$= \sqrt{3} I_\phi \angle -150^\circ$$

$$\mathbf{I}_{cC} = \mathbf{I}_{CA} - \mathbf{I}_{BC} = I_\phi \angle 120^\circ - I_\phi \angle -120^\circ$$
$$= \sqrt{3} I_\phi \angle 90^\circ$$



(a)



(b)

The magnitude of the line currents is √3 times the magnitude of the phase currents and that the set of line currents lags the set of phase currents by 30°.

Example #2

The Y-connected source in *Example #1* (p. 427) feeds a Δ-connected load through a distribution line having an impedance of 0.3 + j0.9 Ω/ϕ. The load impedance is $118.5 + j85.8\ \Omega/\phi$. Use the a-phase internal voltage of the generator as the reference.

- Construct a single-phase equivalent circuit of the three-phase system.
- Calculate the line currents $\mathbf{I}_{aA}$, $\mathbf{I}_{bB}$, and $\mathbf{I}_{cC}$.

a) Calculate the phase voltages at the load terminals.

b) Calculate the phase currents of the load.

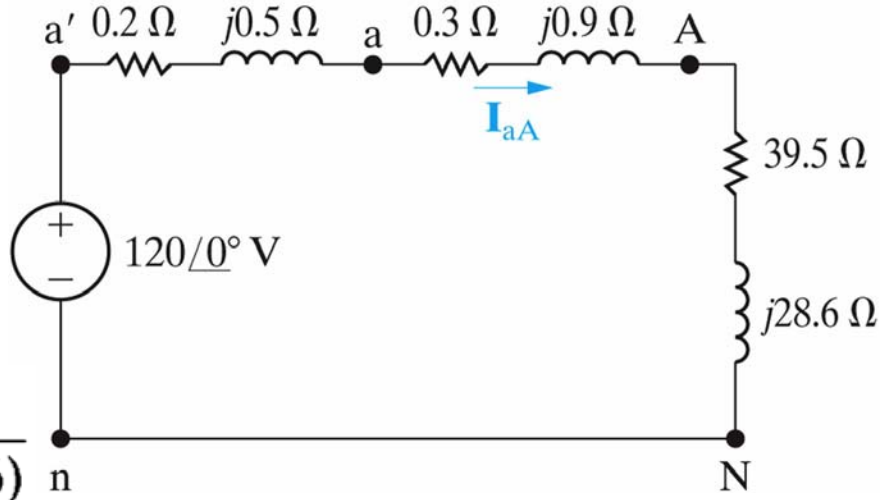
c) Calculate the line voltages at the source terminals.

Solution for Example #2

a)

The load impedance of the Y equivalent is:

$$\frac{118.5 + j85.8}{3} = 39.5 + j28.6\ \Omega/\phi$$



single-phase equivalent circuit

b)

$$\mathbf{I}_{aA} = \frac{120\angle 0^\circ}{(0.2 + 0.3 + 39.5) + j(0.5 + 0.9 + 28.6)}$$

$$= \frac{120\angle 0^\circ}{40 + j30} = 2.4\angle -36.87^\circ\ \text{A}$$

$$\mathbf{I}_{bB} = 2.4\angle -156.87^\circ\ \text{A} \qquad \mathbf{I}_{cC} = 2.4\angle 83.13^\circ\ \text{A}$$

c)

$$\mathbf{V}_{AN} = (39.5 + j28.6)(2.4\angle -36.87^\circ) = 202.72\angle 29.04^\circ\ \text{V}$$

$$\mathbf{V}_{BC} = 202.72\angle -90.96^\circ\ \text{V} \qquad \mathbf{V}_{CA} = 202.72\angle 149.04^\circ\ \text{V}$$

d) The phase currents of the load may be calculated directly from the line currents:

$$\mathbf{I}_{AB} = \left(\frac{1}{\sqrt{3}} \angle 30^\circ\right) \mathbf{I}_{aA} = 1.39 \angle -6.87^\circ \text{ A}$$

$$\mathbf{I}_{BC} = 1.39 \angle -126.87^\circ \text{ A} \quad \mathbf{I}_{CA} = 1.39 \angle 113.13^\circ \text{ A}$$

e)

$$\mathbf{V}_{an} = (39.8 + j29.5)(2.4 \angle -36.87^\circ) = 118.90 \angle -0.32^\circ \text{ V}$$



$$\mathbf{V}_{ab} = (\sqrt{3} \angle 30^\circ)\mathbf{V}_{an} = 205.94 \angle 29.68^\circ \text{ V}$$

$$\mathbf{V}_{bc} = 205.94 \angle -90.32^\circ \text{ V} \quad \mathbf{V}_{ca} = 205.94 \angle 149.68^\circ \text{ V}$$

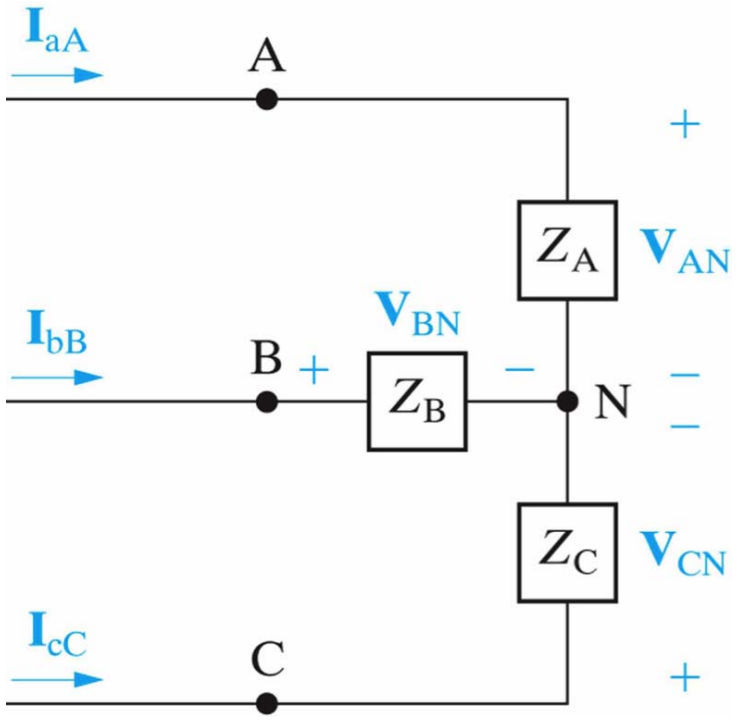
Power Calculations in Balanced Three-Phase Circuits

Average power in a balanced Wye load

$$P_A = |\mathbf{V}_{AN}||\mathbf{I}_{aA}| \cos(\theta_{vA} - \theta_{iA})$$

$$P_B = |\mathbf{V}_{BN}||\mathbf{I}_{bB}| \cos(\theta_{vB} - \theta_{iB})$$

$$P_C = |\mathbf{V}_{CN}||\mathbf{I}_{cC}| \cos(\theta_{vC} - \theta_{iC})$$



Balanced Y load

If we denote

$$V_\phi = |\mathbf{V}_{AN}| = |\mathbf{V}_{BN}| = |\mathbf{V}_{CN}|$$

$$I_\phi = |\mathbf{I}_{aA}| = |\mathbf{I}_{bB}| = |\mathbf{I}_{cC}|$$

$$\theta_\phi = \theta_{vA} - \theta_{iA} = \theta_{vB} - \theta_{iB} = \theta_{vC} - \theta_{iC}$$

We will have

$$P_A = P_B = P_C = P_\phi = V_\phi I_\phi \cos\theta_\phi$$

$$P_T = 3P_\phi = 3V_\phi I_\phi \cos\theta_\phi$$

$$P_T = 3\left(\frac{V_L}{\sqrt{3}}\right) I_L \cos\theta_\phi = \sqrt{3} V_L I_L \cos\theta_\phi$$

V_L and I_L are the rms magnitudes of the line voltage and current

Complex Power in a Balanced Wye Load

Similarly, we can get the following reactive power:

$$Q_\phi = V_\phi I_\phi \sin\theta_\phi$$

$$Q_T = 3Q_\phi = \sqrt{3} V_L I_L \sin\theta_\phi$$

For a balanced load

From the same phase

$$S_\phi = \mathbf{V}_{AN}\mathbf{I}_{aA}^* = \mathbf{V}_{BN}\mathbf{I}_{bB}^* = \mathbf{V}_{CN}\mathbf{I}_{cC}^* = \boxed{\mathbf{V}_\phi \mathbf{I}_\phi^*}$$

Thus, in general

$$S_\phi = P_\phi + jQ_\phi = \mathbf{V}_\phi \mathbf{I}_\phi^*$$

$$S_T = 3S_\phi = \sqrt{3} V_L I_L \angle\theta_\phi^\circ$$

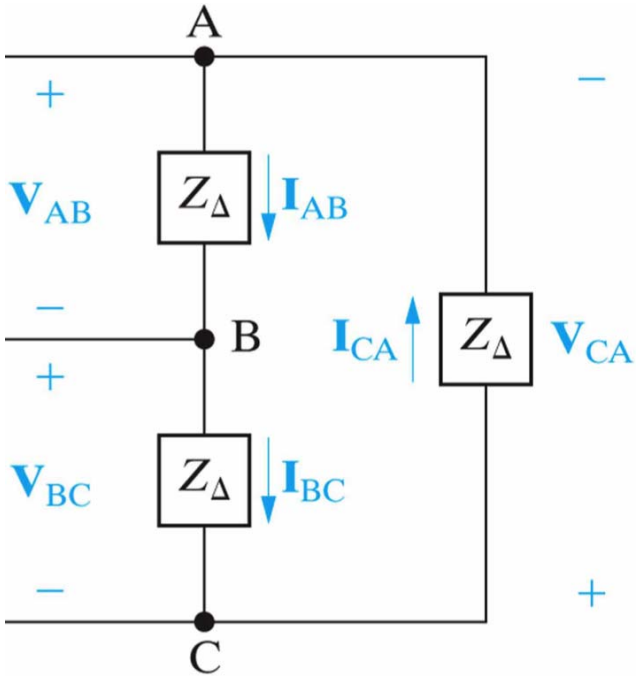
Power Calculations in a Balanced Delta Load

The power associated with each phase is:

$$P_A = |\mathbf{V}_{AB}||\mathbf{I}_{AB}| \cos(\theta_{vAB} - \theta_{iAB})$$

$$P_B = |\mathbf{V}_{BC}||\mathbf{I}_{BC}| \cos(\theta_{vBC} - \theta_{iBC})$$

$$P_C = |\mathbf{V}_{CA}||\mathbf{I}_{CA}| \cos(\theta_{vCA} - \theta_{iCA})$$



For a balanced load,

$$|\mathbf{I}_{AB}| = |\mathbf{I}_{BC}| = |\mathbf{I}_{CA}| = I_\phi$$

$$|\mathbf{V}_{AB}| = |\mathbf{V}_{BC}| = |\mathbf{V}_{CA}| = V_\phi$$

$$\theta_{vAB} - \theta_{iAB} = \theta_{vBC} - \theta_{iBC} = \theta_{vCA} - \theta_{iCA} = \theta_\phi$$

Thus, we have

$$P_A = P_B = P_C = P_\phi = V_\phi I_\phi \cos\theta_\phi$$

Comparing Eq. 11.48 and Eq. 11.34, we can see that, in a balanced load, regardless of whether it is Y- or Δ -connected, *the average power per phase is equal to the product of the rms magnitude of the phase voltage, the rms magnitude of the phase current, and the cosine of the angle between the phase voltage and current*.

The expressions for the total power delivered, reactive power and complex power also have the same form as those developed for the Y load:

$$P_T = 3P_\phi = 3V_\phi I_\phi \cos\theta_\phi = 3V_L\left(\frac{I_L}{\sqrt{3}}\right)\cos\theta_\phi = \sqrt{3}V_L I_L \cos\theta_\phi$$

$$Q_\phi = V_\phi I_\phi \sin\theta_\phi \qquad Q_T = 3Q_\phi = 3V_\phi I_\phi \sin\theta_\phi$$

$$S_\phi = P_\phi + jQ_\phi = \mathbf{V}_\phi \mathbf{I}_\phi^* \qquad S_T = 3S_\phi = \sqrt{3}V_L I_L \angle\theta_\phi$$

Instantaneous Power in Three-Phase Circuits

$$p_A = v_{AN} i_{aA} = V_m I_m \cos \omega t \cos (\omega t - \theta_\phi)$$

$$p_B = v_{BN} i_{bB} = V_m I_m \cos (\omega t - 120^\circ) \cos (\omega t - \theta_\phi - 120^\circ)$$

$$p_C = v_{CN} i_{cC} = V_m I_m \cos (\omega t + 120^\circ) \cos (\omega t - \theta_\phi + 120^\circ)$$

$$V_m = \sqrt{2} V_\phi \quad I_m = \sqrt{2} I_\phi$$

$$p_T = p_A + p_B + p_C = 1.5 V_m I_m \cos \theta_\phi = 3 V_\phi I_\phi \cos \theta_\phi$$

It is invariant with time!

Example #3

a) Calculate the average power per phase delivered to the Y-connected load of Example #1 (P. 427).

b) Calculate the total average power delivered to the load.

c) Calculate the total average power lost in the line.

d) Calculate the total average power lost in the generator.

e) Calculate the total number of magnetizing vars absorbed by the load.

f) Calculate the total complex power delivered by the source.

Solution for Example #3

From Example #1, $V_\phi = 115.22$ V, $I_\phi = 2.4$ A, and $\theta_\phi = -1.19 - (-36.87) = 35.68°$. Therefore

$$P_\phi = (115.22)(2.4)\cos 35.68° = 224.64 \text{ W}$$

The total average power delivered to the load is

$$P_T = 3P_\phi = 673.92 \text{ W}$$

The total power lost in the line is

$$P_{\text{line}} = 3(2.4)^2(0.8) = 13.824 \text{ W}$$

The total internal power lost in the generator is

$$P_{\text{gen}} = 3(2.4)^2(0.2) = 3.456 \text{ W}$$

The total number of magnetizing vars absorbed by the load is

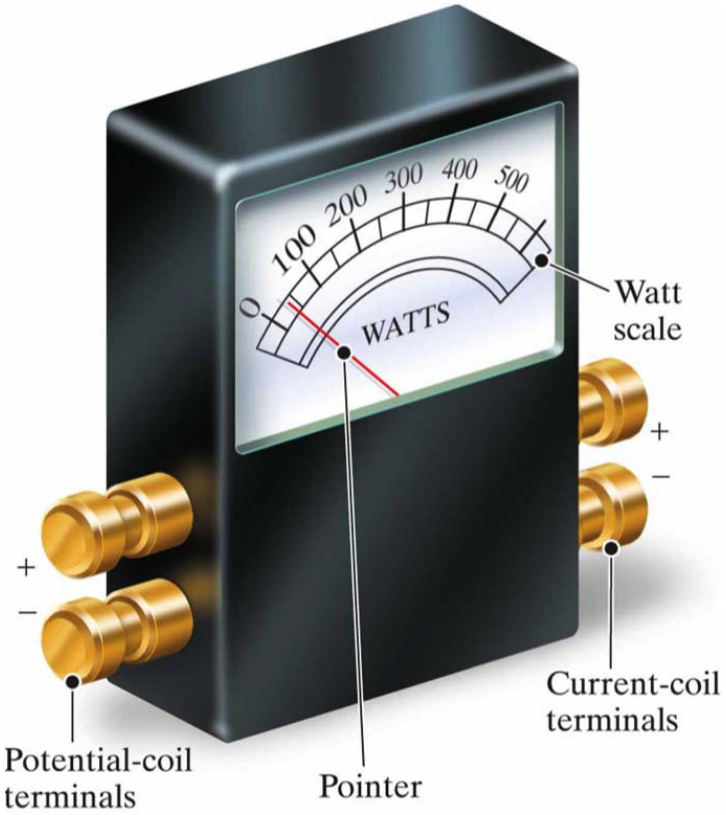
$$Q_T = \sqrt{3}(199.58)(2.4)\sin 35.68° = 483.84 \text{ VAR}$$

The total complex power associated with the source is

$$S_T = 3S_\phi = -3(120)(2.4)\angle 36.87° = -691.20 - j518.40 \text{ VA}$$

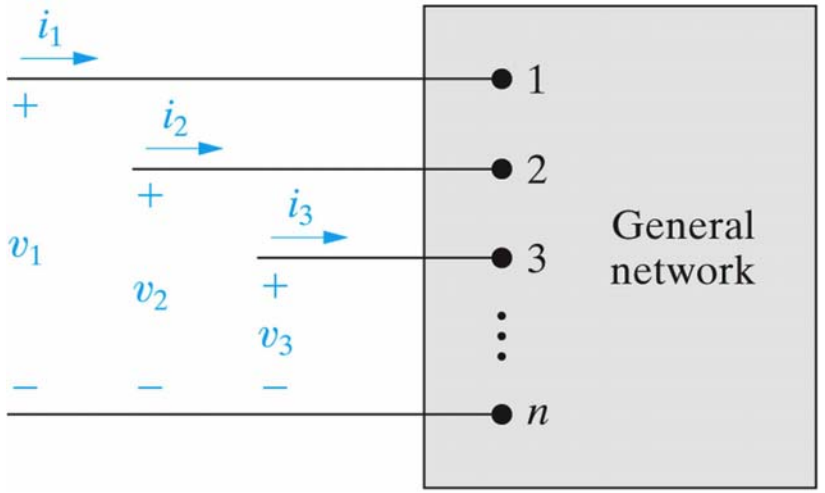
Measuring Average Power in Three-Phase Circuits

The basic instrument used to measure power in three-phase circuits is the electrodynamometer *wattmeter*. It contains two coils. One coil, called the ***current coil,*** *is stationary and is designed to carry a current proportional to the load current*. The second coil, called the ***potential coil,*** *is movable and carries a current proportional to the load voltage*.



The Two-Wattmeter Method

Consider a general network inside a box to which power is supplied by n conducting lines. *If we wish to measure the total power at the terminals of the box, we need to know $n-1$ currents and voltages.* This follows because if we choose one terminal as a reference, there are only $n-1$ independent voltages. Likewise, only $n-1$ independent currents can exist in the n conductors entering the box. Thus the total power is the sum of $n-1$ product terms; that is, $p = v_1 i_1 + v_2 i_2 + \ldots\ldots + v_{n-1} i_{n-1}$.



Applying this general observation, we can see that for a three- conductor circuit, whether balanced or not, we need only two wattmeters to measure the total power.

$$W_1 = |\mathbf{V}_{AB}||\mathbf{I}_{aA}| \cos\theta_1 = V_L I_L \cos\theta_1$$

$$W_2 = |\mathbf{V}_{CB}||\mathbf{I}_{cC}| \cos\theta_2 = V_L I_L \cos\theta_2$$

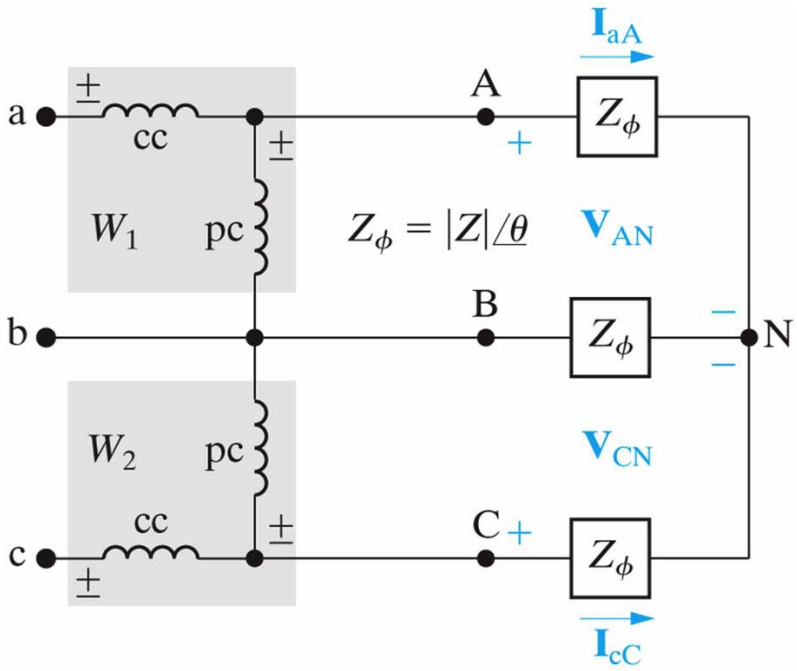
$$\theta_1 = \theta + 30^\circ = \theta_\phi + 30^\circ$$

$$\theta_2 = \theta - 30^\circ = \theta_\phi - 30^\circ$$

$$W_1 = V_L I_L \cos(\theta_\phi + 30^\circ)$$

$$W_2 = V_L I_L \cos(\theta_\phi - 30^\circ)$$

$$P_T = W_1 + W_2 = 2V_L I_L \cos\theta_\phi \cos 30^\circ = \sqrt{3} V_L I_L \cos\theta_\phi$$



1. If the power factor is greater than 0.5, both wattmeters read positive.
2. If the power factor equals 0.5, one wattmeter reads zero.
3. If the power factor is less than 0.5, one wattmeter reads negative。

Conclusion

- Balanced three-phase circuit
- Abc/acb phase sequence
- Y-Y/Y-Δ circuits
- Phase/line voltage/current
- Per-phase average/reactive/complex power
- Two-Wattmeter method